

	COMPASS Field Sampling Protocol Chesapeake Bay Region	
Date Created: 21 Sept 2022 Date Updated: 19 Oct 2022	Creator Name(s): A. Stearns Editor Name(s): A. Stearns	Version: 1
Vegetation Plot Installation and Sampling		

Objective:

To establish replicate vegetation plots that will allow COMPASS to monitor a representative sample of plants and herbaceous ground cover over time.

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I. Experimental Design:

Each Chesapeake Bay Synoptic Site will have five (5) vegetation plots installed in each zone (Wetland, Upland and Transition). These plots will be monitored annually, during peak growth. In the Chesapeake Bay region, peak growth occurs in the transition between the July and August months.

Vegetation plots will be either 1M² or 2M², depending on the zone, and be sectioned with a 10cm PVC grid system.

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Vegetation plots will be annually surveyed using the Braun Blanquet method to determine species density, abundance, and frequency.

Annual vegetation harvests will occur during peak growth. Vegetation from a randomly chosen 10cm grid square from each vegetation plot will be collected, dried and weighed for biomass. Any harvested grid square will not be harvested again for five (5) years.

II. Personal Protective Equipment:

Close-toed shoes and long pants are required at all times when working at COMPASS sites. Garden or work gloves are recommended for hand safety. Eye protection is available and is required in the marsh zones.

III. Installation

A. Materials:

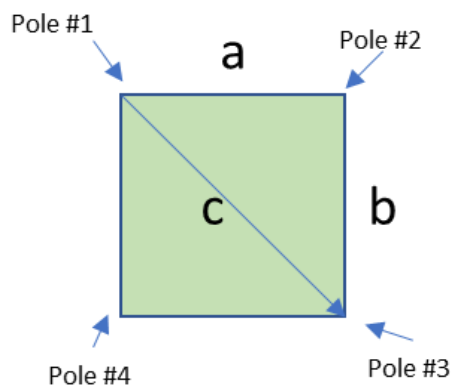
- (3) metric tapes
- Driveway stakes or plastic stemmed flags - 4/plot, 20/zone, or 60/site
- PVC grids
- Vegetation Plot field notebook
- Pencil
- Sharpie

B. Procedure:

1. Designate location of the vegetation plot
2. Set corner pole #1
3. Pull tape in the direction you want to the distance you want the width. *Installing plots to the cardinal direction is always preferred, if possible.*
4. Set corner pole #2

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5. From corner pole #2, pull tape to the distance desired for the length, approximating a right angle, in the proposed direction of pole #3
6. Use the Pythagorean Theorem to find the hypotenuse
Pull tape from pole #1 toward the proposed position of pole #3
7. Place pole #3 where the length of the side intercepts the length of the hypotenuse
8. Pull tapes from pole #'s 1 and 3 to find the placement of the last corner, pole #4
9. Install the PVC grid



$$a^2 + b^2 = c^2$$

$$c = \sqrt{a^2 + b^2}$$

IV. Sampling

A. Materials

- Vegetation Plot field notebook
- Pencil
- Sharpie
- Varying sizes of paper bags (pre-tared bags are recommended)
- Clippers or cutting blade

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- Braun-Blanquet score card

B. Procedure

1. Using the Braun-Blanquet score card, conduct a vegetation survey. Document species scores in the field notebook.
2. Using a random number application, select the 10cm grid square to harvest. If this happens to be a grid that has been harvested in the last five (5) years, select another grid square.
3. Label a paper bag, appropriately sized for the vegetation present.
 - Data
 - Site
 - Zone
 - Plot
 - Grid #
4. Clip the vegetation as close to the ground as possible.
5. Place vegetation in the correctly labeled bag
6. Put samples in the grievie oven at 60°C for at least 72 hours

V. Processing

A. Materials:

- Samples
- Balance
- Weights - 1g and 50g standards
- Weigh boats
- Data collection sheet
- Pencil

B. Procedure:

1. All samples should be dried in the grievie oven at 60°C for at least 72 hours
2. Pull samples out of the oven and let them cool for about 20 minutes
3. Turn on the balance 10 minutes prior to starting
4. Make sure the balance is level

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5. Weigh and record the mass of 1g and 50g standards. Be sure to weigh the standards each new day when using the balance
6. Weigh the sample. If the bag has been pre-tared, you can weigh the sample in the bag documenting the mass of the bag on the data sheet. If not, remove the entire sample from the bag and use a weigh boat to obtain the samples' mass.
7. Record the mass on the data collection sheet

VI. CB Site Installation Logistics

A. Goodwin Islands

- Installed September 2022
- Wetland: 1M² vegetation plots South of the Wetland logger and MET station. Installed along a 30M transect East to west approximately 3M apart from each other. Braun-Blanquet scores obtained. Harvested the NE 10cm grid square for processing.
- Transition: Braun-Blanquet score was obtained from the macrophyte collars. No harvest was done. No additional plots installed for 2022.
- Upland: No plots were installed for 2022

B. Moneystump Swamp

- Installed September 2022
- Wetland: 1M² vegetation plots West of the Wetland logger and MET station. Installed along a 30M transect East to west approximately 3M apart from each other. Braun-Blanquet scores obtained. Harvested the SE 10cm grid square for processing.
- Transition: 2M² vegetation plots were installed. Plot locations were selected using random numbers to select an x,y pace. Braun-Blanquet scores obtained. Harvested the SE 10cm grid square for processing.
- Upland: 2M² vegetation plots were installed. Plot locations were selected using random numbers to select an x,y pace. Braun-Blanquet scores obtained. Harvested the SE 10cm grid square for processing.

C. GCRew

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- Installed October 2022
- Wetland: Vegetation data will be compiled from GCREW's annual plant Census
- Transition: 2M² vegetation plots were installed. Plot locations were selected using random numbers to select an x,y pace. Braun-Blanquet scores obtained. Harvested the SE 10cm grid square for processing.
- Upland: Vegetation data will be compiled from TEMPEST'S bi-annual herbaceous ground cover Census

D. Sweethall Marsh